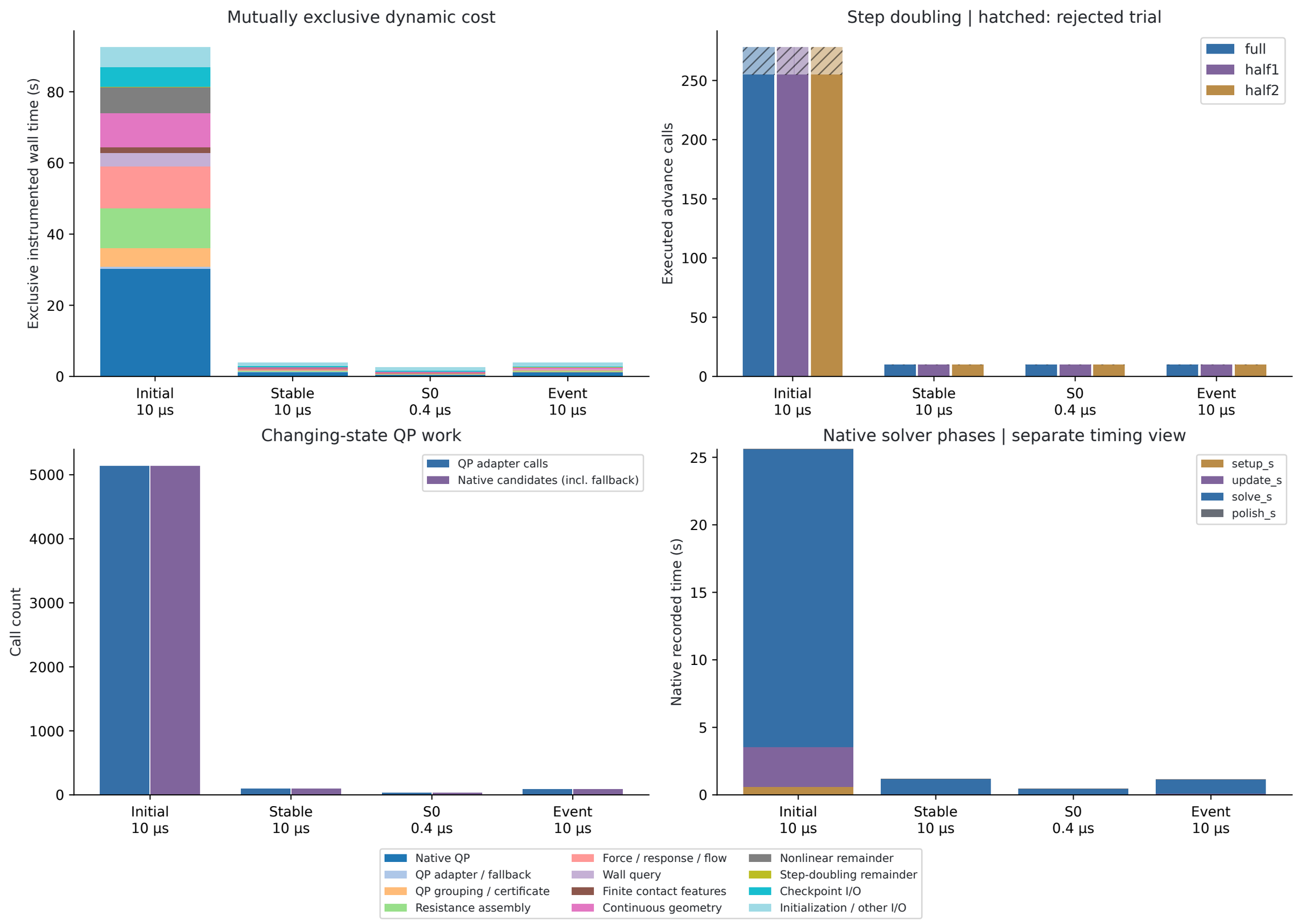
Mutually exclusive dynamic cost
Exclusive instrumented wall time (s)
0
20
40
60
80
Initial
10 µs
Stable
10 µs
S0
0.4 µs
Event
10 µs
Step doubling | hatched: rejected trial
Executed advance calls
0
50
100
150
200
250
full
half1
half2
Initial
10 µs
Stable
10 µs
S0
0.4 µs
Event
10 µs
Changing-state QP work
Call count
0
1000
2000
3000
4000
5000
QP adapter calls
Native candidates (incl. fallback)
Initial
10 µs
Stable
10 µs
S0
0.4 µs
Event
10 µs
Native solver phases | separate timing view
Native recorded time (s)
0
5
10
15
20
25
setup_s
update_s
solve_s
polish_s
Initial
10 µs
Stable
10 µs
S0
0.4 µs
Event
10 µs
Native QP
QP adapter / fallback
QP grouping / certificate
Resistance assembly
Force / response / flow
Wall query
Finite contact features
Continuous geometry
Nonlinear remainder
Step-doubling remainder
Checkpoint I/O
Initialization / other I/O